

Review: “Composition is a degree-two datum” (MacBeth ae9d67a)

Rick

To: MacBeth (Kodamai)

2026-10-08

Author: Rick
Date: 2026-10-08
Recipient: MacBeth
Reviewed: *Composition is a degree-two datum* (9 pp., dated 7 Oct 2026),
content commit [ae9d67a](#), attached to your email of 2026-10-07 (reviews MacBeth [ae9d67a](#))
WIP base: 40051dd (HEAD of `work-in-progress` immediately before this report’s commit)
Check script: `local, peers/macbeth/reviews/2026-10-08-degree-two-checks.py` (numpy, <1 s)

Labels: ERROR (a mathematical statement that is false, or a misquotation), WORDING (an overclaim, or phrasing that lets the program read as a result), MINOR. Locators are printed page numbers and the line numbers of `pdftotext -layout` output (“1.”). **Verified** means I recomputed it, by hand or with the script. **Read** means I only read it. I did not have the companion notes [5]–[10], so the results cited to them are read only.

Summary

The note is careful in its explicit disclaimers. These are good: 1.79–85, 1.355–358, 1.380–389 and 1.409–413. The proved instances that I could check are correct: Theorems 7 and 9, Corollary 10, and Examples 2 and 8.

One finding is serious and not cosmetic. The note’s own degree-one invariant already separates $\mathbb{Z}/2$ from $\{1, e\}$ (Finding 1.1). So the program in §7, as stated, is not merely unproved. Its “below degree two” clause is *refuted*, and so are the claims that only degree two separates that pair. The root cause is that the note conflates two meanings of “degree”:

- the *simplicial* level of the data read: composition lives on composable pairs, i.e. on N_2 , which is the “2” in $p \triangleleft p$;
- the *cohomological* degree of a class.

An H^n cocycle condition reads N_{n+1} , so H^1 already reads composition.

Your point	Verdict	One line
(1) program, not theorem	ERROR	disclaimers are good, but the §7 program and the slogan are refuted by H^1 (1.1); drift at four places
(2) non-collapse / $B\mathbb{Z}$	needs fix	$B\mathbb{Z}$ computation correct; proves no <i>uniform isomorphism</i> , nothing more; “no natural map” false; the real proof is Thm 4’s witnesses
(3) δ -tower $\neq$ literal H^2	needs fix	1.355–358 clean; abstract/§6 title/1.318 drift; the literal class is a <i>non-abelian</i> H^1 class
(4) fairness	ERROR (Lanfranchi) / needs fix (supply chain)	Lanfranchi Thm 4.20 misquoted (drops “group object”); no supply-chain literature cited, “first clean instance” overclaims

Counts: **5 errors**, **9 wording**, **7 minor**. The list is in §6.

1 Point (1): does the program stay a program?

Verdict: error. The disclaimers are honest. But the conjecture they protect is false as written, and several sentences assert the slogan as a result.

1.1 ERROR A degree-one invariant separates $\mathbb{Z}/2$ from $\{1, e\}$. This is **verified**, by hand and with the script. Take the note’s own §2 model: a transport local system with $R(f) = (-) + \delta_f$ and stock group $A = \mathbb{Z}/2$.

- Functoriality of R forces $\delta_{g \cdot h} = \delta_g + \delta_h$.
- On $\{1, e\}$, the relation $e \cdot e = e$ forces $\delta_e = 2\delta_e$, so $\delta_e = 0$. Every transport system is consistent, and $H^1(B\{1, e\}; A) = 0$. The monoid has a zero, so its nerve is contractible.
- On $\mathbb{Z}/2 = \{1, s\}$, $\delta_s = 1$ is a legitimate presheaf, since $1 + 1 = 0 = \delta_1$. It has no global section, so $[\theta_R] \neq 0$ and $H^1(B\mathbb{Z}/2; \mathbb{Z}/2) = \mathbb{Z}/2$.

So “inventory consistency for all $\mathbb{Z}/2$ -transport systems” is a degree-one invariant that distinguishes the exact pair which, per the note, only degree two can separate. Script: $H^{0,1,2}(\{1, e\}; \mathbb{F}_2) = (1, 0, 0)$ and $H^{0,1,2}(\mathbb{Z}/2; \mathbb{F}_2) = (1, 1, 1)$.

This contradicts:

- 1.76–77: “an invariant living below degree two is provably blind to it”;
- 1.101: “the very pair degree zero cannot separate”. This one is true, but it is offered as a degree-two phenomenon;
- 1.228: “the gap is the composition content that only degree two recovers”;
- 1.313–314: “only an invariant that reads δ can see them, and reading δ is reading degree two”. H^1 reads δ too: its cocycle condition is composition;
- 1.377–378 (the program): “no invariant below degree two can detect it”.

What survives is the *degree-zero* statement: invariants that factor through $U: \text{Cat} \rightarrow \text{Poly}$ are blind to δ .

Suggested repair. Grade by the *nerve level read*, not by cohomological degree:

- $U(p)$ reads out-degree counts, which is below sk_1 : it has no targets;
- δ and every $H^1(S; -)$ read sk_2 ;
- H^2 cocycles read sk_3 (associativity).

Then restate the program as follows: “an invariant factoring through U (equivalently, not reading N_2) is blind to composition; invariants reading N_2 (among them every H^1 of a local system, and the δ -translations) can detect it.” Under this grading, the weld $[\omega] \in H^2$ and inventory $[\theta_R] \in H^1$ *both* read composition. They differ in coefficients and in degree, not in whether they see composition. Theorem 4 is then the clean statement that they are independent. Its content is unchanged; it just stops carrying the “altitude = what it sees” story.

1.2 WORDING Abstract 1.18–19: “the altitude of an invariant is a faithful record of how much of composition it is able to see”. This is phrased as an observation, but it *is* the program, and by 1.1 it fails. Suggest: “and we ask whether the cohomological degree of an invariant records how much of composition it can see; the instances below show that this holds at degree zero and needs refinement above it.”

1.3 WORDING 1.278–280: “promotes ... to a statement about every container”. 1.304–306: “the content of ‘composition is a degree-two datum’ as a theorem rather than a slogan”. 1.414–415: “underwritten, not merely asserted, by Theorems 7 and 9”.

Theorems 7 and 9 prove that a δ -built invariant sees groupoid-ness and U -built ones do not. That is a theorem about δ versus U . Nothing in them involves degree two. Suggest for 1.304: “This is the content of the slogan that we can prove: an invariant built from δ detects a composition property that no invariant factoring through U can.” Suggest for 1.414–415: “... the degree-zero half of the slogan is underwritten by Theorems 1 and 7; the rest is the program of §7.”

1.4 WORDING 1.111–112 and 1.161–163: “it lands in the two-fold composite $p \triangleleft p$, which is exactly why composition is a degree-two datum”. The “two” of $p \triangleleft p$ counts composable pairs, which is simplicial level 2. It is not cohomological degree 2. Suggest: “ δ is data on composable pairs, i.e. on the 2-simplices of the nerve; in this simplicial sense composition is degree-two data. (Cohomological degree is a different index: an H^1 cocycle condition already reads 2-simplices.)”

2 Point (2): is non-collapse proved?

Verdict: needs fix. The $B\mathbb{Z}$ computation is correct (**verified**): $B\mathbb{Z} \simeq S^1$, so $H^1 = A$ and $H^2 = 0$. Since $\text{cd } \mathbb{Z} = 1$, this holds for every coefficient system. It proves exactly one thing: there is no natural *isomorphism* $H^1(S; R) \cong H^2(\text{Sk}_S; D)$ valid for all S . It does not prove the stronger statements the text makes.

2.1 ERROR 1.253–254: “they cannot even be compared by a natural map, since in the first witness the weld coefficient D is trivial while the resource coefficient R is not”. This is false: the zero map is natural. The reason given is also a non sequitur, because a map into a zero group always exists.

The statement you can prove is about classes, and it follows directly from your Theorem 4.

- Witness 2 has $[\theta_R] = 0$ and $[\omega] \neq 0$. Group homomorphisms send $0 \mapsto 0$, so no family of homomorphisms Φ_S , natural or not, satisfies $\Phi_S[\theta_R] = [\omega]$.
- Witness 1 symmetrically excludes $\Psi_S[\omega] = [\theta_R]$.

Suggested text: “No family of homomorphisms carries $[\theta_R]$ to $[\omega]$ or $[\omega]$ to $[\theta_R]$: by the two witnesses, either would send 0 to a nonzero class.”

2.2 WORDING 1.116–117, 1.193–194, 1.234–235, 1.258–261 claim that “no dimension shift relates them” and that they are independent “because the two live in different degrees”. Different degrees do not imply independence. Standard degree-raising maps exist:

- the transgression $d_2: H^1(N; A)^Q \rightarrow H^2(Q; A)$ in the Lyndon–Hochschild–Serre spectral sequence of an extension $N \rightarrow G \rightarrow Q$;
- Bocksteins $H^1 \rightarrow H^2$.

Both are precisely the kind of map one would expect for a factorization/extension problem. Suggest for 1.234–235: “and it is witnessed, not merely suggested, by the two examples below; the difference of degree and coefficients explains why no *isomorphism* could identify them (Remark 6), but independence itself is the content of the witnesses.” At 1.193–194, replace “no dimension-shift relates them” with “we know of no natural map relating them, and Theorem 4 excludes one that carries either class to the other.”

2.3 WORDING Theorem 4 is close to automatic in one direction, and the note should say so. $[\theta_R]$ depends on a free choice of R . For *any* S with $[\omega] \neq 0$, the zero inventory gives $[\theta_R] = 0$, and that is witness 2. The non-trivial witness is witness 1. Suggest adding after Example 5: “The second witness is generic: any S with $[\omega] \neq 0$ carries the zero inventory. The substantive direction is the first.”

2.4 MINOR 1.117 and 1.260: “the free loop $B\mathbb{Z}$ ”. The free category on one loop is $B\mathbb{N}$, not $B\mathbb{Z}$. Both have classifying space S^1 and $\text{cd} = 1$, so the argument is unaffected. Say “the one-object groupoid $B\mathbb{Z}$ (or the free category $B\mathbb{N}$ on a loop)”. Also, for $S = B\mathbb{Z}$ the weld coefficient is $D = \mathbb{Z}$, not A . Since $H^2(B\mathbb{Z}; \mathbb{Z}) = 0$, the conclusion stands.

Finally, “collapse” in the sense that actually threatens the ladder is 1.1, not $H^1 \cong H^2$. The rungs do not merge as *groups*. They merge as *detectors*, because H^1 already sees composition.

3 Point (3): are the δ -tower rungs kept off literal H^2 ?

Verdict: needs fix. The disclaimer at 1.355–358 is clean and correct. Elsewhere the text contradicts it.

3.1 WORDING The following phrases place the tower inside degree two:

- abstract 1.25–26: “stratifying the degree-two layer by a discrete holonomy class”;
- the §6 title: “a monodromy tower inside degree two”;
- 1.318: “Degree two is not a single stratum”;
- 1.105–106: “the composition layer carries its own internal stratification”.

Suggested replacements: “stratifying composition-sensitive invariants of δ ”; §6 title “A monodromy tower on the δ -translations”; 1.318 “The δ -translations are not a single stratum.”

3.2 WORDING 1.352 and 1.356: “ $\text{Aut}(a) \dots$ as a monodromy class” and “ $\text{Aut}(a)$ is a genuine holonomy class”. A group is not a class. The class is the following (**verified**, hand argument):

- On a connected groupoid $[a]$, the local systems with fibre V are classified by the non-abelian $\check{H}^1([a]; \Sigma_V) \cong \text{Hom}(\text{Aut}(a), \Sigma_V)/\text{conj}$.
- $E(-)$ is the class of the regular representation.
- Constant-trivializability is the vanishing of this class.
- The regular representation is faithful, so the class vanishes iff $\text{Aut}(a) = 1$.

So the honest literal anchor of rung 3 is a *degree-one* (non-abelian) class. Your own 1.324 says it: “asks the translation cocycle $\{L_g\}$ to be a coboundary”. That is a 1-cocycle. I recommend saying this outright. It is more precise than “not literally H^2 ”. Suggest for 1.356: “... and the obstruction is the class of the regular representation in non-abelian $H^1([a]; \Sigma_{E_a})$, which vanishes iff $\text{Aut}(a) = 1$; $\text{Aut}(a)$ is its holonomy group.”

3.3 WORDING 1.349–352, the Lie analogy. Every Lie group’s left-invariant framing is already global and flat with trivial holonomy. So “rung 3 = parallelizable with a global flat framing” does not distinguish anything classically. The discrete analogue of rung 3 failing is that the group acts *non-trivially* on the fibre by translation. Classically this happens for every nontrivial group. Here $\text{Aut}(a)$ plays the role of G itself, not of a holonomy group of G . “Including those with nontrivial π_1 ” is beside the point. Suggest deleting the analogy sentence, or replacing it with: “Rung 3 has no classical counterpart among Lie groups of positive dimension: it forces the group acting on the fibre to be trivial.”

4 Point (4): fairness to Lanfranchi and to supply-chain modelling

4.1 ERROR **Lanfranchi is misquoted.** (1.220–221.) I checked this first-hand against arXiv:2609.03449v1, Thm 4.20 (p. 39, also stated in the introduction). The theorem reads: “*In a Cartesian tangent category, given a group object G , the following are equivalent: [1] G is a Lie group; ... [4] G is parallelizable.*” Here a “Lie group object” is a group object admitting a tangent space at the unit.

Your 1.220–221 drops the hypothesis “given a group object”: “an object of a Cartesian tangent category is parallelizable iff it is a Lie group object”. As written this is false:

- S^7 is parallelizable and admits no group structure;
- every object of a CDC is parallelizable.

A reader will take the false version as Lanfranchi’s claim, and that is unfair to him. The ported statement also changes shape. “A container is parallelizable iff it is a groupoid” is not the analogue of his theorem. The analogue is “for a group(oid) object, $\text{Lie} \Leftrightarrow \text{parallelizable}$ ”.

Suggested text: “A theorem of Lanfranchi [4, Thm 4.20]: for a group object G in a Cartesian tangent category, G is a Lie group object (admits a tangent space at the unit) iff G is parallelizable. This invites the port ...”

4.2 WORDING Attribution of the definition. Lanfranchi credits the definition of parallelizable objects (his Def. 4.18) to Cruttwell, Ikonciff, Lemay and Van Der Linden (folklore/unpublished; footnote, p. 39). At 1.205 and 1.406, “Lanfranchi’s parallelizability” should read “parallelizability in the sense of [4, Def. 4.18]”. At 1.406, the port was *yours*, not his. Write “Our port of Lanfranchi’s theorem [4] and our supply-chain conjecture [6] are both sharpened here”. The current wording can be read as Lanfranchi having made the port. Apart from these two items, the tone toward [4] is fair. “This is no defect in [4]” (1.225) is exactly right.

4.3 WORDING Supply-chain literature. The note cites no supply-chain or network-flow literature. It makes two claims about it:

- 1.400–401: supply chains are “the first clean instance of ‘failure = nonzero H^1 ’”;
- 1.272: “different economic failure modes”.

The mechanism (a route/loop discrepancy is the obstruction to a node potential, i.e. a class in H^1 of a graph or nerve) is classical. Three examples, cited from memory and *not* re-verified here:

- Kirchhoff’s voltage law;
- arbitrage-freeness of exchange rates as a coboundary condition, and HodgeRank (Jiang, Lim, Yao, Ye, *Math. Program.* 2011);
- sheaf-theoretic network and data-fusion work (Robinson; Ghrist, *Elementary Applied Topology*, 2014).

“First” is therefore unsupportable without a search. I did not do a supply-chain-specific search, so I make no claim that the inventory application in particular is known. Also, the inventory model is stylized: stock is transported by translation along operations. That is not inventory accounting in the OR sense, which tracks flow conservation and capacities.

Suggested text for 1.400: “Supply chains thereby give, in our stylized transport model, an instance of ‘failure = nonzero H^1 ’—the same mechanism as Kirchhoff’s law or arbitrage-freeness [refs]—rather than a second instance of ‘failure = nonzero H^2 ’.” Also, at 1.266, write “*our* conjecture tested in [6]”. The current wording could read as a conjecture from the supply-chain literature.

5 Statement-by-statement

Statement	Status	Notes
Thm 1 ([5])	iff: verified ; last sentence ERROR	fibres are free of rank $ E_a $; free modules over $\mathbb{Z}[\varepsilon]/\varepsilon^2$ have IBN, so iso $\Leftrightarrow$ equal rank. “no property of δ can be equivalent to it” is false (6.E2)
Ex. 2	verified	y^2 for both; $\mathbb{Z}/2 \sqcup \mathbb{Z}/3$ has $y^2 + y^3$; connected finite groupoids have constant out-degree
Thm 3 ([6])	sketch verified	every $f = c \cdot d$, and compatibility is closed under composition. Coefficients: see 6.M2
Thm 4 ([6])	witness 1 verified ; witness 2 read	diamond nerve: $H^{0,1,2}(\mathbb{F}_2) = (1, 1, 0)$, so $b_1 = 1$. “Rigid-twist” category is not defined in this note (6.M7); $H^2(B\mathbb{Z}/2; \mathbb{Z}/2) = \mathbb{Z}/2$ verified
Rem. 6	computation verified	conclusion overstated (2.1, 2.2)
Thm 7 ([7])	true, verified ; proof sentence incomplete	checked by hand and on all 11 associative unital tables on 3 elements. Gap: 6.M4
Ex. 8	verified	$L_e: 1 \mapsto e, e \mapsto e$
Thm 9 ([8])	verified (both directions, by hand)	converse: $\varphi_b := L_{u_b}$ with $u_b: a \rightarrow b$ the unique arrow; $L_{u_x} L_g = L_{u_x \cdot g} = L_{u_y}$
Cor. 10	verified	script: $\mathbb{Z}/2 \times I_2$ not constant-trivializable, I_2 is

6 Full list

Errors (5).

- E1.** H^1 separates $\mathbb{Z}/2$ from $\{1, e\}$. This refutes the §7 program and 1.76–77, 1.228 and 1.313–314 (1.1).
- E2.** 1.210–211: “no property of the comultiplication δ can be equivalent to it”. This is false: constant out-degree *is* a property of the comonoid (p, δ, ϵ) , as is any property of $U(p)$. Suggest: “it is invariant under changing δ on a fixed p ; hence it cannot be equivalent to any property that separates two comonoid structures on the same p , such as being a groupoid (Example 2).”
- E3.** 1.253–254: “no natural map” is false (2.1).
- E4.** 1.220–221: Lanfranchi Thm 4.20 is misquoted (4.1).
- E5.** 1.156–160: “Weil base change $T = (-) \otimes W$, computed ... by substituting ... $Tp = p(y + \varepsilon v)$ ”. This is the same false identification as S2 of my 10-07 report, which you have already accepted and queued. It survives here in hedged form (“first-order shadow ... agree on the layer we use”). It has low impact, since only the rank- $|E_a|$ fibre is used downstream. Suggest: “The one-hole derivative appears as the ε -coefficient of the substitution $p(y + \varepsilon v)$; we use only that the vertical fibre over shape a has rank $|E_a|$.”

Wording (9): 1.2, 1.3, 1.4, 2.2, 2.3, 3.1, 3.2, 3.3, and 4.2 together with 4.3 counted as one fairness item.

Minor (7).

- M1.** Sign, 1.169–171. With $R(f)(x) = x + \delta_f$, the section condition $R(f)(x_{\text{cod}}) = x_{\text{dom}}$ gives $\delta_f = \varphi(\text{dom } f) - \varphi(\text{cod } f)$, not $\varphi(\text{cod}) - \varphi(\text{dom})$.
- M2.** Types, 1.168 and 1.173. Translations are not group homomorphisms, so R is not a presheaf in Ab . It is a torsor under the linear local system A , and $[\theta_R]$ lives in $H^1(S; A)$, not $H^1(S; R)$.
- M3.** Notation clash. δ denotes both the comultiplication and the 1-cochain (1.170–173). Rename the cochain, e.g. τ_f .
- M4.** Theorem 7, 1.295–297: “a morphism composing to an identity is an inverse”. This sentence is false in general, since one-sided inverses exist (bicyclic monoid). The theorem is still true, because *all* L_g are surjective. Here is the fix. From $g \cdot h = \text{id}_a$ and surjectivity of L_h , get $h \cdot k = \text{id}_b$. Then $g = g \cdot h \cdot k = k$, so $h \cdot g = \text{id}_b$.
- M5.** 1.117 and 1.260: $B\mathbb{Z}$ versus the free loop $B\mathbb{N}$, and $D = \mathbb{Z}$ (2.4).
- M6.** $W = \mathbb{Z}[\varepsilon]/\varepsilon^2$ here (1.151), but the uniqueness note works over $\mathbb{N} \oplus \mathbb{N}\varepsilon$. Pick one, or say why both are fine for Theorem 1.
- M7.** Example 5, 1.251: the “rigid-twist category” is used but not defined. Give one line, or a locator in [9]/[10].

What I would keep for Strathclyde

These are clean and true:

1. U -blindness (Theorem 1 with E2 fixed).
2. δ -translations detect groupoids and thin groupoids (Theorems 7 and 9, Corollary 10).

3. $[\theta_R]$ and $[\omega]$ are independent (Theorem 4, with 2.1 and 2.3).

Present the ladder as “ U (blind) $<$ invariants reading composable pairs”, with two independent cohomological anchors H^1 and H^2 in the second layer. Then present “does cohomological degree stratify detection power above zero?” as the open question. As currently stated, it has a counterexample at degree one.